



Klein bottles

Cliff Stoll makes all kinds of Klein bottles, which are a very impractical way of storing liquids. <https://dgit.in/jan20-09>



Flat cosmos

All the evidence from various instruments that points to us living in a universe with a flat curvature. <https://dgit.in/jan20-10>

SHAPE OF THE UNIVERSE

We have an incomplete picture of cosmic geometry but scientists have made some really smart guesses.

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ne of the earliest cosmological questions to puzzle mankind, was whether the universe was finite or infinite.

Throughout history, the scientific consensus seems to have swung between the two possibilities. Ancient Greek philosophers were divided on the issue, but the atomists primarily believed that the universe was infinite. However, the most influential philosophers including Plato, Aristotle and Ptolemy believed that the universe was finite. Cosmas Indicopleustes, a Greek monk argued that “The crystal-made sky

sustains the heat of the Sun, the moon, and the infinite number of stars, otherwise, it would have been full of fire, and it could melt or set on fire.” If you ignore the bit about the crystal sky, this is an important observation as it makes the point that a universe with infinite stars would be on fire. In Medieval Europe, Copernicus and Kepler believed in a finite universe. Descartes and Newton, who were influential figures in classical cosmology believed in an infinite universe. Einstein argued for both a finite as well as infinite universe (though not at the same time). Around the turn of the twentieth century, a paradox had emerged, named after the German astronomer, Heinrich Wilhelm Olbers. If the universe was infinite, then it should have been full of light, uniform in all

directions. There were some early proposed resolutions to the paradox. One was that there were a finite number of stars in an infinite universe. Another was that interstellar dust was obscuring the light from the stars. These are not really resolutions, because given infinite time, then the dust would also start radiating heat, and a finite number of stars in an infinite universe should still be enough to light up the sky in the night, as well as the day. The only possible resolution was that the universe is expanding so we cannot see stars beyond a certain limit, and that it has a particular age. As far as current science can tell, the last two are both true. Considering the limitations of the speed of light, the observable universe is a sphere centered around the Earth, with a radius of 46.5 billion